

Modular Lunar Settlement Concept Design for Long-term Habitation Located in Lava Tube

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Abstract

"Space Architecture Workshop, Moon Village Edition" was held at the Department of Architecture at Xi'an Jiaotong-Liverpool University in December 2023. This paper presents one of the projects proposed by a team of 3 students, which aims to design a sustainable and expandable lunar settlement for long-term human staying. The design strategy consists of placing most of the settlement in a lava tube to gain natural protection, using prefabricated and inflatable fan-shaped modular units to promote transportation efficiency and utilisation flexibility. The proposed Moon Village has a sustainable self-sufficient system, and the internal functional layout is arranged based on the astronauts' daily schedule. The Moon Village will initially expand from surface to underground, where the water resource may be located. In the future, it can be extended horizontally along the lava tubes and developed into a lunar underground city.

Keywords: Moon village; Lava tube settlement; Modular Units; Sustainable development; Expandable modules

1. Introduction

1.1 Background of Space Exploration

From the beginning of the Space Race in the middle of the 20th century, humanity has persistently striven to extend the frontiers of its existence beyond the planet. During this process, the Moon is one of the locations where humans attempt to establish the first extraterrestrial presence. The vision of a Moon settlement was nurtured in 2016 by Jan Woerner [1], the director general of the European Space Agency, spurring global interest and creative energy by bringing humans to the Moon to stay. Establishing a lunar settlement has also been embraced by the Chinese Space Agency, which has developed some design schemes such as Xuanwu Station lunar habitation [2], "Clover" and "Red Star" [3].

Many studies have demonstrated that the Moon has some favourable survival conditions for terrestrial living beings. In particular, several Space exploration teams reported the presence of water as ice in its underground [4], [5]. Besides, lava tubes could provide natural shielding for settlement [6], while several potential lunar subsurface openings have been observed by radar evidence [7].

In the last decades, there have been some achievements in dealing with hostile lunar contexts and resource shortages. The environmental control and life support system in the International Space Station [8], the four-biological cycle system implemented at China's Lunar Palace 1 [9], and the Biosphere II project life system [10] provided bases for the lunar settlement's sustainable staying. These achievements contribute to human survival and long-term activities on the Moon in the future. During the process of extending human presence beyond Earth, it will be necessary to have an interdisciplinary approach, by involving also experts from architecture and building disciplines to enhance the user experience, allowing more sustained Space exploration [11].

This paper introduces a possible lunar settlement design based on the existing literature theories. It starts with analysing some key factors that will affect staying in the Lunar context, formulating the research questions, proposing a design solution and then demonstrating the project's future development and conclusion.

1.2 Lunar context

The huge environmental difference between Earth and the Moon represents a significant challenge for

long-term humans staying. According to Naser and Chen [12], the key challenges are as follows: firstly, the lack of atmospheric conditions on the Moon makes long-term human activities an issue. Secondly, it is essential to determine how to circumvent the potential impact of meteorites on the lunar surface and the elevated levels of cosmic radiation. Additionally, extreme temperatures and daily temperature fluctuations pose significant challenges. Furthermore, it is necessary to investigate differences between lunar gravity and the terrestrial one, which may affect the construction, production, and survival conditions. It is important to identify strategies to mitigate the detrimental effects of these aspects.

2. Design focus

Considering the lunar conditions as a design context, the present paper investigates a hypothetical design for Moon Village, focusing on the following aspects:

- (1) Where would be the ideal location for the initial stages of a lunar settlement?
- (2) What characteristics should the settlement possess to ensure long-term staying and consider inhabitants' daily needs/comfort?
- (3) How can the lunar settlement be built and expanded according to future needs?

3. Design concept

The design focuses on the settlement's appropriate locations, construction methods, and self-sufficient development strategies. In the present research, three key aspects have been developed: the underground construction process, modular units, and sustainability. These principles collectively aim to facilitate the establishment of a permanently habitable lunar base.

3.1 Underground

Compared to the lunar surface environment, the subterranean environment offers protection from cosmic radiation, meteorite impacts, and temperature fluctuations, which are hostile to long-term human survival and building feasibility [13]. The hypothetical location for the construction of the lunar village has been selected in a potential lava tube situated within the perpetual day zone of the Moon in the south polar region. Studies conducted by the Politecnico di Torino in 2021 [14] and Mogilever *et al.* [15] have demonstrated that constructing lunar habitats within lava tubes is a viable and contemporary approach. Additionally, scholars have proposed a range of potential scenarios, including the concept of habitats within pressurised lava tubes, as outlined by Martin and Benaroy [6].

3.2 Modular units

The design incorporates prefabricated and inflatable modules to facilitate the Moon Village's construction and flexible development. Additionally, due to the lower radiation exposure and more liveable temperatures underground, the building envelope requires lower strength than a lunar surface construction, making it suitable to use soft, inflatable materials. The advantages of prefabricated and inflatable modules are both in the transport and construction processes since the inflatable units significantly reduce the cargo volume after compression, thereby reducing the transportation cost and improving efficiency. Reaching the destination, the modules can be deployed with breathable air and reorganised according to functional needs, which could be different during the stages of development.

3.3 Sustainability

Due to the resource shortage and cargo transport limitation, the design proposes a self-sufficient development strategy focusing on local energy production by employing solar panels and food production from a self-cycling lunar greenhouse. The design utilises solar energy resources, including light for underground illumination and useful radiation for energy production. Meanwhile, the greenhouse's four-biological cycle system, a closed integrative bio-regenerative life support system through integrating efficient plant cultivation, animal protein production, and bioconversion of solid waste, will be beneficial to achieving the long-term development goal.

4. Results

The research results are reflected in the design and plans for the Moon Village. The following section will present the design results in three specific areas: I) construction process, II) functional programming, and III) sustainability principles.

4.1 Construction process

The construction of the lunar base can be divided into three periods (Fig. 1). Namely, the preparation period (stage 0), the development period (stage 1), and the expansion period (stage 2).

4.1.1 Stage 0

The preliminary phase consists of site surveys and preparation carried out by robots, and transporting components plus other construction materials from Earth by rockets. The launch vehicle will arrive on the lunar surface carrying the components in batches.

In Step 0.1, the lunar robots will arrive for site exploration and preparation. After surveying the lava tube and determining its stability, the robots will use lunar soil to build temporary shelters on the surface next to the tubes, where materials and equipment will be

stored. In Step 0.2, based on the site survey and analysis, the lava tube will be reinforced to ensure the safety of subsequent construction.

Then, in Step 0.3, the settlement's components, which consist of the vertical transportation core modules and the components of the lunar surface greenhouse, will be delivered. The vertical core will be inserted into the lava tube, and a 300-degree sector platform will be 3D printed around the core to hold modular units, leaving a 60-degree gap to provide space for the transport platform. The transport platform will be able to move vertically and will be placed along the core cylinder to enable module allocation in the next step. During this phase, the robots will continuously build the greenhouse on the surface.

This phase ends with the completion of 1) the core cylinder connecting to the underground space; 2) the fixed platforms hosting the settlement modules; 3) the movable vertical transfer platform; and 4) the greenhouse.

4.1.2 Stage 1

The development period consists of modules'

increment in the platforms and the vertical extension since they can reach different levels through the movable platform.

Considering the different lighting requirements of the functional areas and the settlement development stages, the design envisages an evolving layout for the Moon settlement at various stages. In step 1.1, the settlement is designed to accommodate 8 people. The residential modules, storage and technical modules, work modules and public modules will all be located on the first-floor underground platform while receiving natural light to enable underground spaces to mimic Earth's daytime environment, allowing staff to maintain rhythms of life [16]. In step 1.2, it is expected to accommodate approximately 30 people, and the residential modules will increase correspondingly. The work, storage and technical modules will be partially moved to the lower platform and replaced by the residential and public modules on the first one because the upper floor has better daylight. In step 1.3, the Moon Village will accommodate about 80 people. The three platforms will have a clear functional focus (Fig. 3). All the residential modules are concentrated on the B1 floor

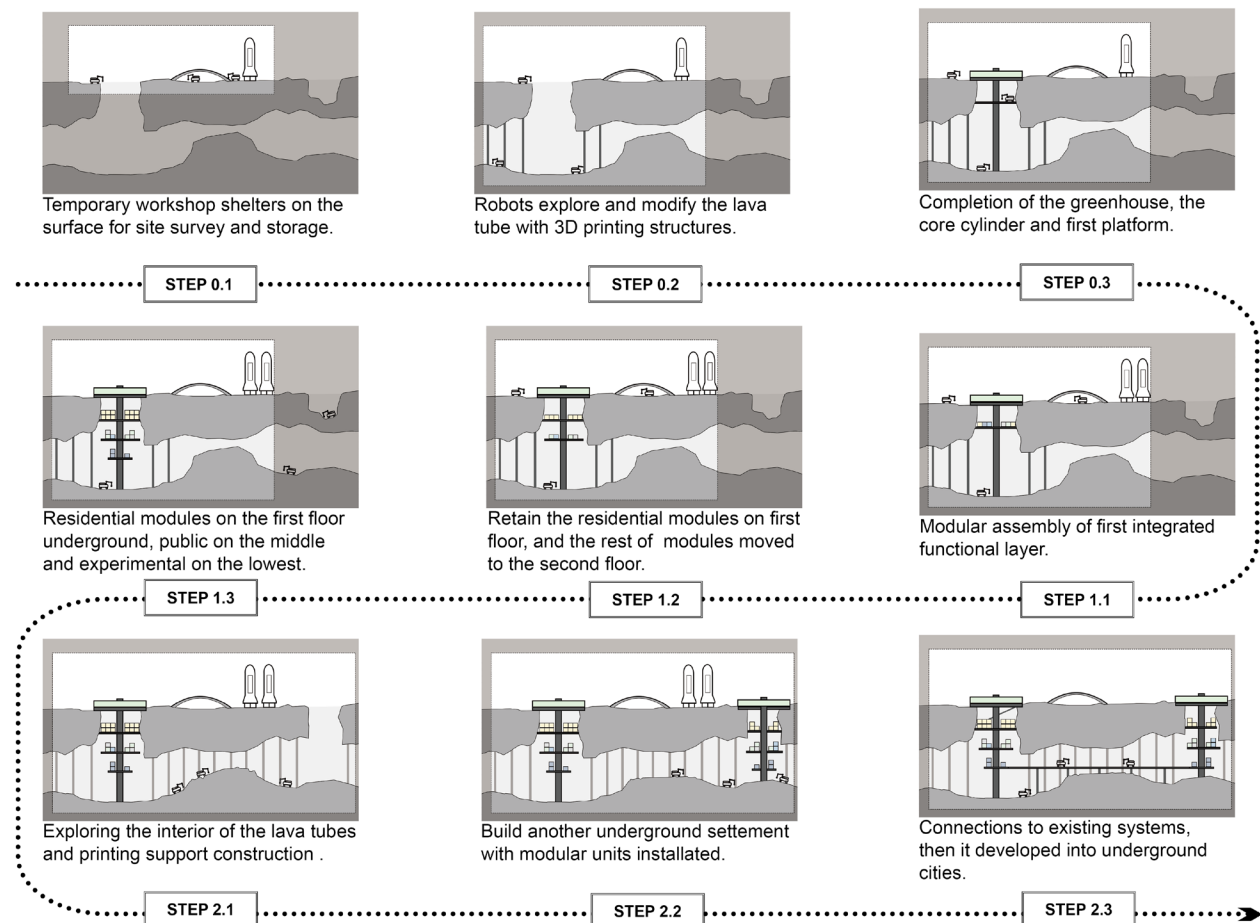


Fig. 1. Construction Process

platform. The work, storage and technical modules are mainly on the B2 floor. Labs for lava tubes and water ice will be put in the lowest basement level, while the movable platform can transfer goods and modules vertically through different levels.

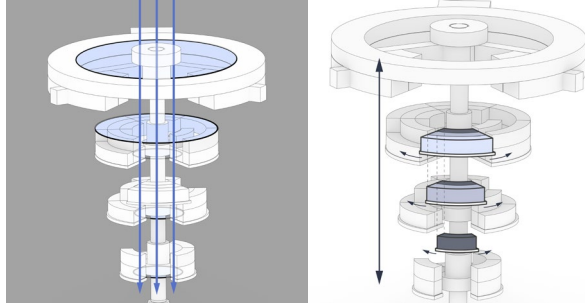


Fig. 2. Arrange the units according to their light requirement and the moveable platform transfers vertically and rotates around the core

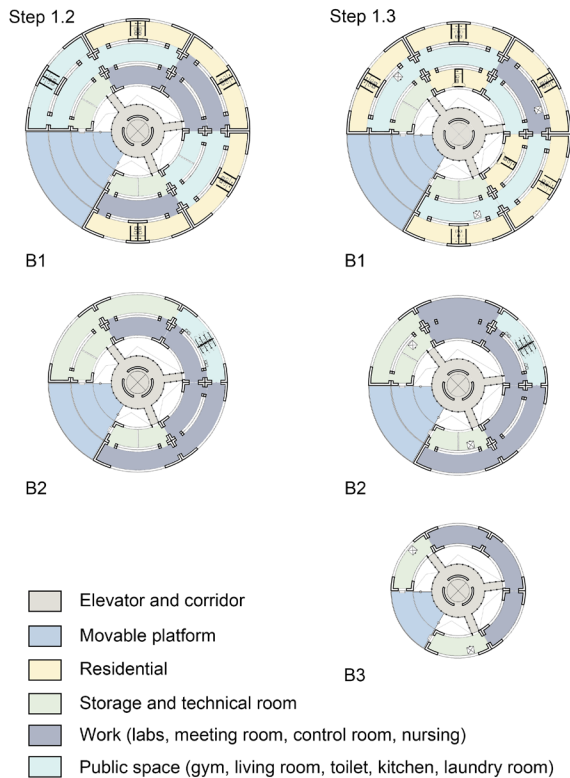


Fig. 3. Step1.2 and 1.3 functional changing

4.1.2 Stage 2

In Stage 2 (steps 2.1-2.3), robots will explore the adjacent lava tubes for resources, including lunar soil and water ice. During this process, the Moon settlement may be horizontally expanded by connecting nearby lava tubes or develop along the lava tube to form a larger-scale interconnected system.

4.2 Functional programming

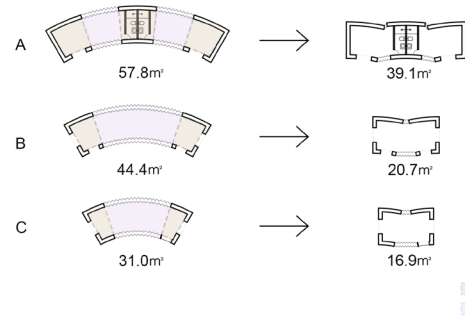
Based on the study of the astronauts' daily activities [17], the Moon Village was designed with prefabricated and inflatable modular units which could be used for different functions.

4.2.1 Functional units

The modular units are characterised by both prefabrication and inflatability. The advantages are as follows:

Firstly, due to the current limitations of the lunar industry, the prefabrication of modules on Earth guarantees precise manufacturing for the initial development of a lunar base.

Secondly, the inflatable feature increases the efficiency of transporting modules. The deployed area will be reduced significantly since the space between the rigid compartment is enclosed by inflatable walls that can be compressed (Fig. 4). Fig. 5 shows the number of rockets needed to transport modules for the underground part by using Space X Starship at different stages.



Unit Type	Rigid Area (S_{Ra})	Flex Area (S_{Fa})	Deployed Area (S_{Da})	Compressed Area (S_{Ca})	Reduced Percentage $[(S_{Da}-S_{Ca})/S_{Da}]$
A	33.6m ²	24.2m ²	57.8m ²	39.1m ²	31.4%
B	16.2m ²	28.2m ²	44.4m ²	20.7m ²	53.4%
C	10.8m ²	20.2m ²	31.0m ²	16.9m ²	45.5%

Fig. 4. Units compression

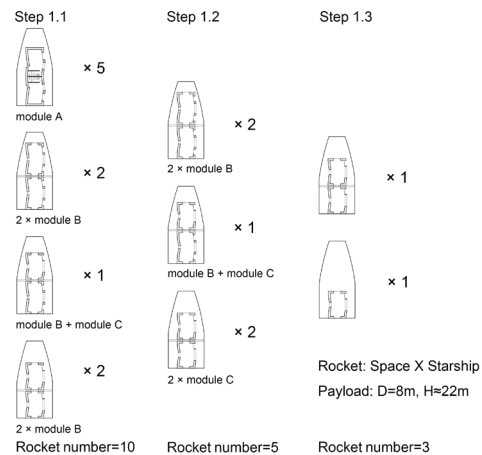


Fig. 5. Rockets number needed for step 1

Thirdly, since the settlement required a closed, breathable environment with oxygen, air inflation inside the units could achieve it.

In addition, the modules are built to uniform specifications and can be transformed into different functional units by simply changing the interior configurations and furnishings. Such characteristics make it easier to change and replace individual units at different stages of their functionality and expand, with more room for change. The internal arrangement can be automated by the method proposed by Zhu [18], considering internal connectivity [19].

As a result, the utilisation of prefabricated and pre-integrated units achieves benefits in terms of lower costs and less labour work. The remaining construction tasks, consisting of the construction of lava tube support and base skeleton, as well as the repair of units, can be assigned to multi-functional robots and reduce human work in extreme conditions.

4.2.2 Units programming

On each platform, there are three different sizes of fan-shaped modular units that form a 60-degree sector, which can be divided into the service space (rigid compartment) and the serviced space (inflatable compartment) as shown in Fig. 6. A pre-integrated bathroom block is located in the middle of the outer module. Equipment such as life-support devices are placed inside rigid compartments composed by ceilings, walls and floors while providing space as a corridor. By combining the units and putting different functions and furniture inside, a fully functional floor will be formed (Fig. 8). While the mobile platform transports materials vertically, the fan-shaped units could rotate around the elevator along the rail on each level to the movable platform, which will carry them to different floors according to Moon Village's development stages.

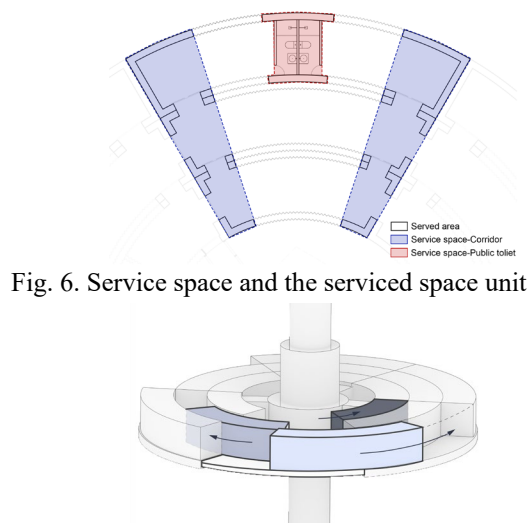
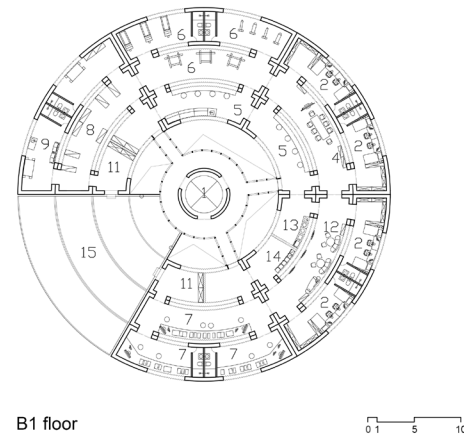


Fig. 6. Service space and the serviced space unit

Fig. 7. Units rotate through the rail on the platform



- | | | |
|-----------------|----------------------|-----------------|
| 1. Elevator | 9. Nursing | 17. Reception |
| 2. Residential | 10. Canteen | 18. Garden |
| 3. Toilet | 11. Airlock | 19. Greenhouse |
| 4. Meeting room | 12. Living room | 20. Office area |
| 5. Lab | 13. Kitchen | |
| 6. Gym | 14. Laundry room | |
| 7. Control room | 15. Movable platform | |
| 8. Storage | 16. Technical room | |

Fig. 8. Step 1.1 B1 plan

As mentioned, the settlement can accommodate 8 astronauts in its first stage, with the functional layout of the plan designed according to the astronauts' daily schedule [17]. The bedrooms are close to the public living room, where they can have meals, and the next-door meeting room is used to arrange work tasks for the whole day. After that, astronauts can start daily physical exercise and work in the corresponding area. In particular, these inflatable walls are translucent and can integrate windows to introduce natural light, which to some extent may improve the users' experience [16].

4.3 Sustainability

With long-term development as the primary goal, sustainability is a key factor for the Moon Village. Due to significant environmental differences, the Moon lacks many necessary resources for human survival, such as accessible water, food, and energy. The cost of transporting supplies, for instance, construction materials, from Earth to the Moon will be extremely high. In order to achieve a sustainable long stay, in-situ resource utilization (ISRU) and self-sufficient strategies are applied to this design.

4.3.1 ISRU

The resources that have been considered for the construction and the functioning of the Moon Village include solar energy resources, lunar soil resources, and

possible water ice resources in craters in the permanently shadowed region of the Moon, specifically in the Antarctic region [5].

Solar energy is captured by the photovoltaic panels in the greenhouse. Concurrently, the sunlight of high-intensity radiation that goes through the translucent membrane in the greenhouse significantly diminishes its harmful radiation, providing illumination for the personnel inside the lava tubes. The sunlight also imitates Earth's environment, thereby assisting the crew in regulating their biological rhythms.

The abundant lunar soil within the lava tubes and on the lunar surface is employed principally for research purposes and to provide the raw materials for the settlement building using a 3D printing strategy. According to Ellery and Alex [20], various metals, ceramics, and volatiles can be extracted from lunar minerals to facilitate the construction of a lunar base.

In addition, the self-sufficient biological system in the present design concept will partly rely on potential icy water that can be extracted and obtained on the Moon, as suggested by many studies [5]. Moreover, Jin *et al.* [4] have provided evidence that some lunar soil contains hydrated minerals $(\text{NH}_4)\text{MgCl}_3 \cdot 6\text{H}_2\text{O}$, which include water.

4.3.2 Self-sufficient system

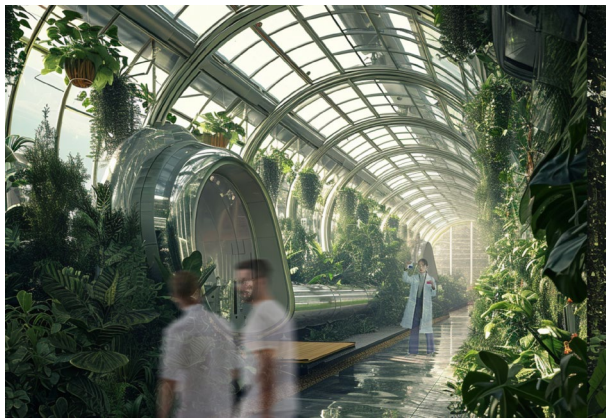


Fig. 9. Greenhouse render

On the lunar surface, the greenhouse is designed to make the settlement more self-sufficient. Except for material supply from Earth to the surface greenhouse and underground units in the early stages, the whole circulatory system is mainly between the lunar surface greenhouse and underground units. The crews' wastewater and exhaled carbon dioxide will be recycled and treated in a waste disposal system and then transit carbon dioxide and water to the greenhouse, where plants will convert them into oxygen and provide food. Here, the biomass produced by organisms through the cleaning treatment system can also be used to feed

yellow mealworms and mushrooms, which provide nutrients like protein for crews [21].

Additionally, the lunar surface greenhouse not only guarantees the material circulation and energy supply for crews but also provides a natural garden-like psychological healing space for the crews at the same time.

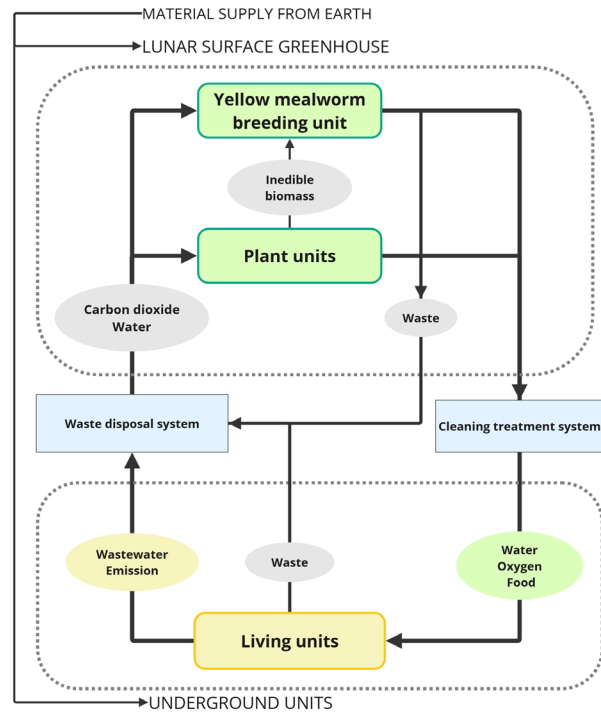


Fig. 10. Substance circulation

5. Future development

The present project provides an initial design for a Moon Village, but some limitations remain.

Firstly, the greenhouse is not gradually expanded together with the increase in personnel at different stages of development. Instead, a relatively large scale was built from the beginning, which resulted in huge construction costs in the initial stage. Also, whether the plant greenhouse can perform a fully self-sufficient system requires more rigorous calculations. Secondly, the design requires lava tubes with certain dimensional compatibility in terms of size and shape, and there is limited possibility of looking further into the subsurface by current technology as radar discovery, mainly due to geometric and solar illumination constraints [7]. Thirdly, when it expands along the lava tube, it can't be guaranteed that every newly added settlement has a natural light source. Searching for the skylight formed by natural collapse may result in two "settlements" being far apart, reducing their direct accessibility to each other. Another way is manually excavating skylights, but it may disrupt the stability of the lava tube.

The above shortcomings are hoped to be gradually improved in future research.

Conclusion

In summary, this paper presents a Moon Village design based on existing research and it focuses on workflow for the construction process, modular units, and sustainability. After analysing the lunar environment, it was decided to place the lunar settlement inside a lava tube at the Moon's south pole. The Moon base will initially be expanded downwards from the skylight on the lunar surface and eventually expanded laterally along lava tubes to form an underground moon city. The internal flow of Moon Village is arranged reasonably according to the daily schedule of astronauts. The utilisation of in-situ resources and the attainment of partial resource self-sufficiency are the main methods to achieve sustainability.

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References

- [1] 'Moon Village'. Accessed: Aug. 11, 2024. [Online]. Available: https://www.esa.int/About_Us/Ministerial_Council_2016/Moon_Village
- [2] 'In-situ construction method for lunar habitation: Chinese Super Mason', *Autom. Constr.*, vol. 104, pp. 66–79, Aug. 2019, doi: 10.1016/j.autcon.2019.03.024.
- [3] 'Research on Building Plans Design for Future China Lunar Base', *J. Deep Space Explor.*, vol. 9, no. 6, Art. no. 6, Feb. 2023, doi: 10.15982/j.issn.2096-9287.2022.20210061.
- [4] S. Jin *et al.*, 'Evidence of a hydrated mineral enriched in water and ammonium molecules in the Chang'e-5 lunar sample', *Nat. Astron.*, pp. 1–11, 2024.
- [5] T. Chakraborty, T. H. Syed, E. Heggy, D. Putrevu, and U. Dutta, 'On the reachability and genesis of water ice on the Moon', *ISPRS J. Photogramm. Remote Sens.*, vol. 211, pp. 392–405, May 2024, doi: 10.1016/j.isprsjprs.2024.03.020.
- [6] R. P. Martin and H. Benaroya, 'Pressurized lunar lava tubes for habitation', *Acta Astronaut.*, vol. 204, pp. 157–174, Mar. 2023, doi: 10.1016/j.actaastro.2022.12.013.
- [7] L. Carrer, R. Pozzobon, F. Sauro, D. Castelletti, G. W. Patterson, and L. Bruzzzone, 'Radar evidence of an accessible cave conduit on the Moon below the Mare Tranquillitatis pit', *Nat. Astron.*, pp. 1–8, Jul. 2024, doi: 10.1038/s41550-024-02302-y.
- [8] 'NASA Achieves Water Recovery Milestone on International Space Station - NASA'. Accessed: Aug. 12, 2024. [Online]. Available: <https://www.nasa.gov/missions/station/iss-research/nasa-achieves-water-recovery-milestone-on-international-space-station/>
- [9] Y. Fu, R. Guo, and H. Liu, 'An optimized 4-day diet meal plan for "Lunar Palace 1"', *J. Sci. Food Agric.*, vol. 99, no. 2, pp. 696–702, Jan. 2019, doi: 10.1002/jsfa.9234.
- [10] M. Nelson, P. B. Hawes, and M. Augustine, 'Life systems for a lunar base', Sep. 1992. Accessed: Aug. 02, 2024. [Online]. Available: <https://ntrs.nasa.gov/citations/19930004804>
- [11] Z. Zhu, A. Fioravanti, and U. Coraglia, 'Space Vehicle-Building Design Process Issues and Models. A framework', Nov. 2020.
- [12] M. Z. Naser and Q. Chen, 'Extraterrestrial Construction in Lunar and Martian Environments', in *Earth and Space 2021*, Virtual Conference: American Society of Civil Engineers, Apr. 2021, pp. 1200–1207. doi: 10.1061/9780784483374.111.
- [13] A. K. Theinat *et al.*, 'Lunar lava tubes: Morphology to structural stability', *Icarus*, vol. 338, p. 113442, Mar. 2020, doi: 10.1016/j.icarus.2019.113442.
- [14] L. Mainini and R. Landolina, 'Study and development of a small lunar habitat inside Moon Lava Tubes based on ARTEMIS model', 2021, Accessed: Jul. 18, 2024. [Online]. Available: <https://webthesis.biblio.polito.it/20950/1/tesi.pdf>
- [15] N. B. Mogilever *et al.*, 'Expedition Cognition: A Review and Prospective of Subterranean Neuroscience With Spaceflight Applications', *Front. Hum. Neurosci.*, vol. 12, p. 407, Oct. 2018, doi: 10.3389/fnhum.2018.00407.
- [16] 'The Human Factor in the Settlement of the Moon', [Online]. Available: https://kns.cnki.net/kcms2/article/abstract?v=_NV5_o307BybLvI5Poe5d6ODwW7z7rNaj3itjTyAODI3HbkBtRciMOMwCBS2O3s8MCxqx2bZ2cotVwqjZxoL2aXfoqzT1FSuFT2VB3WbbZrvIV34YnIZiw-0VSMzMibQMvYgrbFv7mV4ERlCIHBxJQ==&uniplatform=NZKPT&language=gb
- [17] M. Simon, A. Whitmire, C. Otto, and D. Neubek, 'Factors impacting habitable volume requirements: Results from the 2011 Habitable Volume Workshop', 2011. Accessed: Aug. 09, 2024. [Online]. Available: <https://ntrs.nasa.gov/citations/20110023287>
- [18] Z. Zhu, U. Coraglia, and A. Fioravanti, 'PASTA: Parametric Approach for Space Transplanet hAbitations - A generative powered design process for internal configurations in spaceships', Jan. 2022, pp. 47–56. doi: 10.52842/conf.ecaade.2022.2.047.

[19] Z. Zhu, ‘SAUCE – SpAcevehicle-bUilding Connectivity Evaluation: An Adjacent Matrix Based Digital Tool for Internal Connectivity Evaluation and Improvement in Spacevehicle-Buildings’, 2024, pp. 85–92. doi: 10.1007/978-981-97-0621-1_11.

[20] A. Ellery, ‘Leveraging in situ resources for lunar base construction’, *Can. J. Civ. Eng.*, vol. 49, no. 5, pp. 657–674, May 2022, doi: 10.1139/cjce-2021-0098.

[21] Y. Fu, R. Guo, and H. Liu, ‘An optimized 4-day diet meal plan for “Lunar Palace 1”’, *J. Sci. Food Agric.*, vol. 99, no. 2, pp. 696–702, Jan. 2019, doi: 10.1002/jsfa.9234.

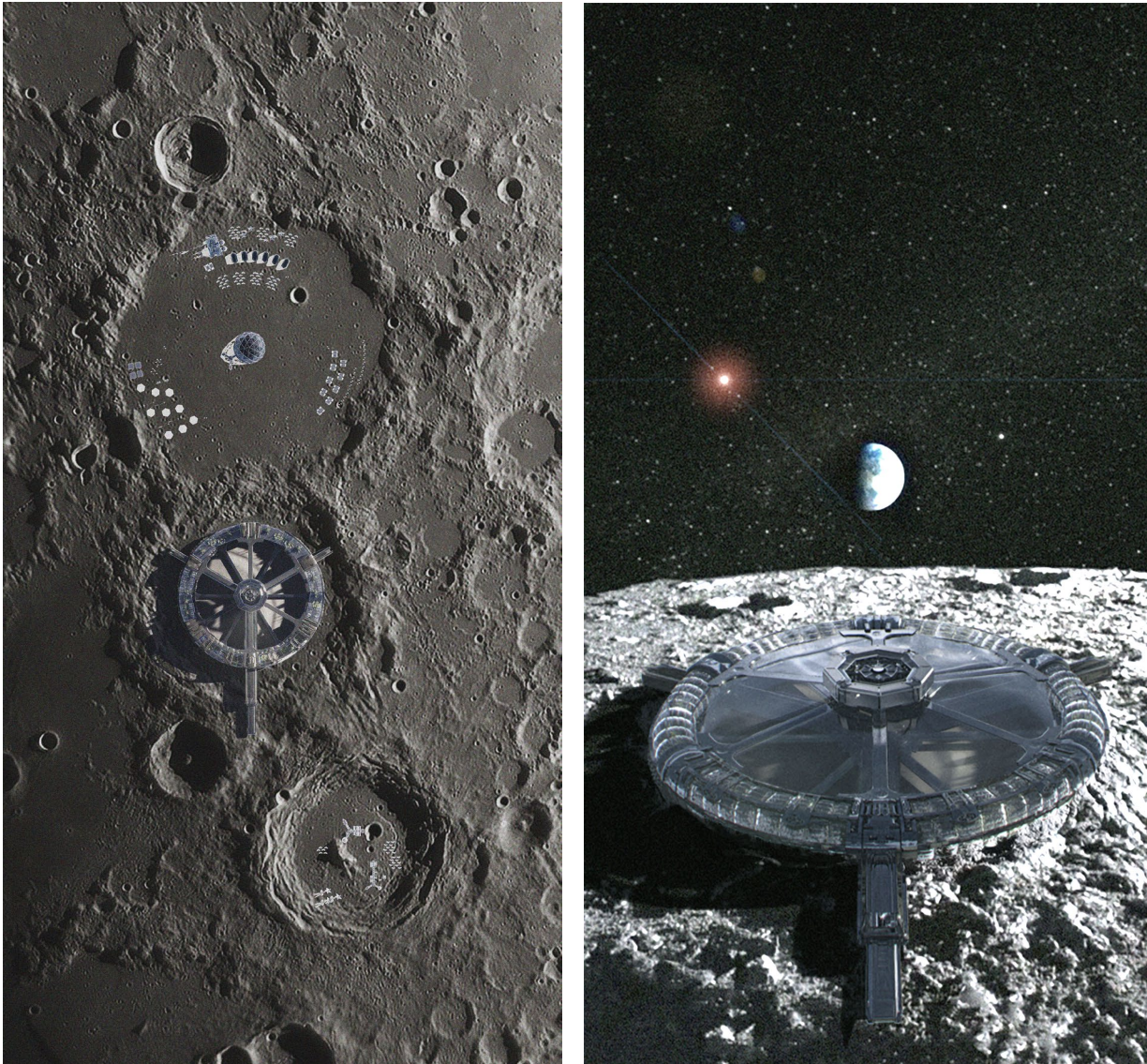


Fig. 11. Render: Aerial view of Moon Village



Fig. 12. Render: Underground units